AHAD HOSEN AKASH

B.Sc. in Computer Science and Engineering | Machine Learning Intern
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Links: LinkedIn | GitHub | ORCID | Medium.

RESEARCH PROFILE

Computer Science and Engineering graduate (CGPA 3.71, 5th in department) with research experience in NLP and graph machine learning, clinical machine learning, network security, and edge-based IoT systems, alongside hands-on full-stack and mobile development. A consistent methodological focus on leakage-safe evaluation, benchmark integrity, and reproducibility. One co-authored manuscript under review, two first-author manuscripts in preparation, and currently exploring agentic AI combined with blockchain. Seeking a funded Master's or research assistant position.

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science and Engineering *2026*
Jamalpur Science and Technology University (JSTU), Bangladesh
CGPA: **3.71 / 4.00** (First Class, Fifth Position)

Higher Secondary Certificate (HSC), Science *2020*
Birshreshtha Munshi Abdur Rouf Public College (BMRPC), Bangladesh
GPA: **5.00 / 5.00**

Secondary School Certificate (SSC), Science *2017*
Zeal Bangla Sugar Mills High School, Bangladesh
GPA: **5.00 / 5.00**

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Machine Learning Intern (Remote) June *2026 - Present*
FlyRank AI

- Contribute to model development, evaluation, and documentation within a professional engineering workflow.

Enumerator (Contract) *November 2025 - December 2025*
Islamic Relief Bangladesh

- Participated in the Endline Survey of the Safe-WASH Project in Jamalpur Zilla, focusing on data collection methodologies, digital survey tools, ethical guidelines, and quality assurance to ensure accurate and reliable baseline data for the project.

Enumerator (Contract) June 2025 - July 2025
Islamic Relief Bangladesh

- Participated in the Baseline Survey of the Safe-WASH Project in Dewangonj Upazila, focusing on data collection methodologies, digital survey tools, ethical guidelines, and quality assurance to ensure accurate and reliable baseline data for the project.

RESEARCH EXPERIENCE

Fake Review Detection with a Hybrid DeBERTaV3 + GraphSAGE Model *(undergraduate research)*

- Designed a hybrid text-and-graph architecture combining DeBERTaV3 text encodings with GraphSAGE over a 359K-node Yelp review graph; reached 0.9951 AUC on known entities.

- Audited the pipeline for metric-to-code alignment and corrected methodology errors (encoder freezing, pooling strategy, token truncation, aggregation type) so that the manuscript matches the implementation.

Proactive Personal Safety via Edge-Driven Machine Learning *(co-author)*

- Contributed to a dual-device ESP32 IoT framework (wearable Safety Unit and Guardian Unit) with on-device logistic regression risk inference, GPS/GSM hardware, and a direct-to-tower SOS fallback.
- Reported ROC-AUC 0.941, PR-AUC 0.962, 22.58 µs inference latency, and 5-7 s SOS delivery.
- Under review at *Results in Engineering*.

PUBLICATIONS

Under review

1. S. Bhattacharjee, M. A. Ihsan, A. S. Mahmud, M. N. Al Islam (corresponding), M. K. U. Ahamed, S. Roy, and **A. H. Akash**. "Proactive Personal Safety via Edge-Driven Machine Learning: An Embedded IoT Risk Awareness Framework." Submitted to *Results in Engineering* (Manuscript RINENG-D-26-13932).

PROJECTS

BMMS: Bangladesh Mess Management System | Django REST Framework, React, TypeScript, PostgreSQL
Multi-tenant web app for shared student and bachelor households, covering meals, bazaar duty rotation, expenses, monthly bills, payments (cash, bKash, Nagad, bank), inventory, and attendance. Modular Django REST backend with role-based memberships (Owner, Manager, Member) across multiple messes per user, Google OAuth, and an immutable audit log. Data isolation enforced with PostgreSQL Row-Level Security; deployed on Supabase, Render, and Vercel.
github.com/AHAAkash/-Bangladesh-Mess-Management-System

Krishi Bondhu | Google AI Agents Intensive ("Agents for Good" track)
Multi-agent AI advisor for Bangladeshi smallholder farmers (crop diagnosis, weather, market guidance) built with Google ADK, FastMCP, and Gemini Flash models, with hard-coded safety guardrails against banned pesticides. github.com/AHAAkash/krishi-bondhu

MonBondhu | Bilingual (English and Bangla) mental health companion app
Flutter (Dart), SQLite, Google Gemini. Offline-first app with AI chat, mood and sleep tracking, a private journal, emergency helplines, and a doctor directory; Gemini powers context-aware chat, journal risk analysis, personalized routines, and weekly insights.
github.com/AHAAkash/MonBondhu-Mental-Health-App

My_ChatGPT | AI-powered conversational assistant
TypeScript, JavaScript, CSS, Gemini API. An AI chat application designed to support natural-language conversations, answer questions, and assist with everyday tasks using conversational AI.
github.com/AHAAkash/My_ChatGPT

TEACHING AND TECHNICAL WRITING

- **Private Tutor**
  Self-Employed | January 2023 – August 2026
  - Provided one-to-one academic support to students in relevant subjects.
  - **Tutored 100+ students**, adapting teaching methods to their individual learning needs and academic goals.
- Technical articles on Claude Code and AI frameworks, published on Medium.

AWARDS AND ACHIEVEMENTS

- Participant, Google AI Agents Intensive (Kaggle) and "Agents for Good" Hackathon
- Second Position, Idea Contest of CSE Association (2024)
- First place, Upazila-level Mathematics Olympiad (2015)
- First place, Upazila-level ICT Olympiad (2016)

SKILLS

Research and ML: Machine Learning, Deep Learning, NLP, graph neural networks (GraphSAGE), transformer models (DeBERTaV3), gradient boosting (CatBoost), SHAP interpretability, nested cross-validation, leakage auditing, TRIPOD+AI reporting

Programming: Python, Java, JavaScript, TypeScript, Dart (Flutter), C, C++, SQL

Libraries and tools: Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Jupyter, LaTeX, Git/GitHub, GitLab, CI/CD pipelines, Linux

Software development: HTML, CSS, FastAPI Framework, REST API design, JWT authentication, PostgreSQL/Prisma, MySQL, SQLite, Google Gemini API, Google ADK, Firebase, Supabase

Vibe Coding Tools: ChatGPT, Claude, Grok, Perplexity AI, DeepSeek, Gemini, Cursor, Github Copilot, Google Antigravity

CS fundamentals: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, System Design, Cryptography, Networking

Soft skills: Scientific writing, research reproducibility, documentation, team collaboration, public speaking

COMMUNITY ENGAGEMENT AND ACTIVITIES

Formal Executive Secretary,
CSE Association, Jamalpur Science and Technology University
October 2024 - March 2026

Science Club member,
Birshreshtha Munshi Abdur Rouf Public College, Bangladesh
October 2017 - January 2019

Former Boy Scout Leader
Zeal Bangla Sugar Mills High School, Bangladesh
March 2014 – December 2016

REFEREES

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